VIETNAM NATIONAL UNIVERSITY
HCMC
INTERNATIONAL UNIVERSITY
School of Biotechnology
Department of Biotechnology

COURSE SYLLABUS

Course Name: BIOSTATISTICS

Course Code: **BT317IU**

1. General information

Course designation	*The course introduces students to the scope, nature, tool and interpretation of elementary statistics: descriptive statistics, inferential statistics, hypotheses testing methods, and regression.*
Semester(s) in which the course is taught	3
Person responsible for the course	Nguyen Tan Khoi Vo Thi Minh Thu
Language	English
Relation to curriculum	*Compulsory*
Teaching methods	*lecture, lesson.*
Workload (incl.contact hours, self-study hours)	*(Estimated) Total workload: 85* *hContact hours: lecture: 25* *hours* *Private study including examination preparation, specified in hours[9]: 60 hours*
Credit points	2 credits (theory) ~ (3.1 ECTS)
Required and recommended prerequisites for joining the course	Calculus I, II

[9] When calculating contact time, each contact hour is counted as a full hour because the organization of the schedule, moving from room to room, and individual questions to lecturers after the class, all mean that about 60 minutes shouldbe counted.

<table>
<tr><td>Course objectives</td><td>

Be able to perform standard statistical tests with common statistical packages
Respect different opinions
Understanding (of other's needs)
Active in finding solutions for challenges or problems
Enhance students' critical thinking in domains involving judgments based on data and stimulate the type of independent thinking requiring research beyond the confines of the textbook, through projects, interdisciplinary examples and exercises

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<tr><td>Course learning outcomes</td><td>Upon the successful completion of this course students will be able to:
<table>
<tr><th>Competency level</th><th>Course learning outcome (CLO)</th></tr>
<tr><td>Knowledge</td><td>CLO1. Learn to understand the main features of statistics; Understand the role of statistics and data analytic methods. Understand the fundamental principles of statistical inference
CLO2. Be able to distinguish the general principles underlying the most common tests</td></tr>
<tr><td>Skill</td><td>CLO3. Be able to analyze statistical data properly; Know the assumptions of common tests andimpacts of violations</td></tr>
<tr><td>Attitude</td><td>CLO4. Active in finding solutions for challenges or problems
Be able to solve practical problems and analyze thesolutions</td></tr>
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<tr><td>Content</td><td>
Lecture 1: Basic concepts (CLO1, CLO2)
- Populations, unit, observations, samples
- Sampling, random samplings
- Sample parameters
- Descriptive and inferential statistics
Processing data and presenting data
- Scale of measurement; Frequency table
- Histogram, frequency polygon, dot plot, bar graph, pie graph, frequency curve, scatter diagram
Sample parameters:
Central Tendency: mean, median, mode.
Variability: standard deviation, variance, degree of freedom, coefficient of variation

Lecture 2: (CLO1, CLO3)
Probability: Frequency; definitions of probability; rules for probability calculation.
Probability distribution, discrete
- Binomial
- Poisson
Lecture 3: Continuous distribution of probability: (CLO1, CLO3)
- normal
- standardized normal
Continuous distribution of probability:
- t-distribution
- chi-square distribution
- F-distribution.

Lecture 4: Estimation of parameters (CLO1, CLO2, CLO3)
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	- sample mean - population proportion - sample variance Central Limit TheoremInterval of confidence **Lecture 5: Hypothesis testing (CLO1, CLO2)** - Principles - Experimental design - Hypothesis/conditionsComparing means - t-test - F-test - testing population proportion. **Lecture 6: ANOVA (CLO1, CLO2)** - one way ANOVA - two-way ANOVA - Tukey test, LSD test. **Lecture 7: Non-parametric tests. (CLO1, CLO2)** **Lecture 8: Regression analysis. (CLO1, CLO2)**
Examination forms	*Written exams (Midterm + Final)*
Study and examination requirements	Students are expected to spend at least 8 hours per week studying this course. This time should be made up of reading, working on exercises and problems, group assignment and attending class lectures and tutorials. University regulations indicate that students may be refused to take final assessment if they attend less than 80% of scheduled classes. Regular attendance is essential for successful performance and learning in this course, particular in the view of the interactive teaching and learning approach adopted.
Reading list	Textbooks: Jim Fowler, Lou Cohen, and Ph. Jarvis. 1998. Practical Statistics for Field Biology. John & Wiley & Sons. 2nd Edition. References: Chap T. Le. 2003. Introductory Biostatistics. John & Wiley & Sons. J. Susan Milton and Jesse C. Arnold. 2003. Introduction to Probability and Statistics. Mc Graw-Hill. 4th Edition.

2. Learning Outcomes Matrix (optional)

The relationship between Course Learning Outcomes (CLO) (1-5) and Program/Student Learning Outcomes (PLO) (1 - 8) is shown in the following table:

	PLOs							
CLO	1	2	3	4	5	6	7	8
1		X						
2		X						
3							X	
4						X		

**Use Bloom's Taxonomy*

3. Planned learning activities and teaching methods

Week	Topic	CLO	Assessments	Learning activities	Resources
	Lecture 1: Basic concepts - Populations, units, observations, samples - Sampling, random samplings - Sample parameters - Descriptive and inferential statistics Processing data and presenting data - Scale of measurement; Frequency table - Histogram, frequency polygon, dot plot, bar graph, pie graph, frequency curve, scatter diagram Sample parameters: Central Tendency: mean, median, mode	CLO1, CLO2	10%	- Lecture presentation - Group forming - Class discussion Read book &lecture 1	
2, 3	Lecture 2: Sample parameters (cont): Variability: standard deviation, variance, degree of freedom, coefficient of variation Probability: Frequency; definitions of probability;rules for probability calculation. Probability distribution, discrete - Binomial - Poisson	CLO1, CLO3	10%	- Lecture presentation - Class discussion - Read book &lecture 2.	
4	Lecture 3: Continuous distribution of probability: - normal standardized normal Continuous distribution of probability: - t-distribution - chi-square distribution - F-distribution.	CLO1, CLO3	10%	- Lecture presentation - Class discussion Read book &lecture 3.	
5	Lecture 4: Estimation of parameters - Sample mean - Population proportion - Sample variance Central Limit Theorem Interval of Confidence	CLO1, CLO2, CLO3	15%	- Class discussion - Read book &lecture 4.	
6, 7	Lecture 5: Hypothesis testing - Principles - Experimental design	CLO1, CLO2	15%	- Lecture presentation - Class discussion - Read book &lecture	

	- Hypothesis/conditions Comparing means - t-test - F-test - Testing population proportion			5	
8	**Midterm**				
9	Lecture 6: ANOVA - One-way ANOVA - Two-way ANOVA - Tukey test, LSD test	CLO1, CLO2	15%	- Lecture presentation - Class discussion Read book &lecture 6	
10	Lecture 7: Non-parametric test	CLO1, CLO2	15%	- Lecture presentation - Class discussion Read book &lecture 7	
11	Lecture 8: Regression analysis	CLO1, CLO2	10%	- Lecture presentation - Class discussion - Read book &lecture 8	
17	**Final exam**				

4. Assessment plan

Assessment Type	CLO1	CLO2	CLO3	CLO4
Class participation (10%)	60%Pass	60%Pass		
Quiz (15%)	Quiz 1 50%Pass	Quiz 2 50%Pass	Quiz 1, 2 50%Pass	
Midterm (25%)			50%Pass	50%Pass
Final exam (50%)			50%Pass	50%Pass

Note: %Pass: Target that % of students having scores greater than 50 out of 100.

5. Rubrics

5.1. Holistic rubric

Holistic rubric for evaluating the entire document, e.g., exercises/quizzes/HW	
Score	**scription**
5	monstrates complete understanding of the problem. All requirements of task are included in response
4	monstrates considerable understanding of the problem. All requirements of task are included
3	monstrates partial understanding of the problem. Most requirements of task are included.
2	monstrates little understanding of the problem. Many requirements of task are missing.
1	monstrates no understanding of the problem.
0	response/task not attempted

Note: this rubric is also used to evaluate questions in an exam.

5.2. Analytic rubric

2. Critical thinking value rubric for evaluating questions in exams:

	Capstone	Milestone		Benchmark
	4	3	2	1
Explanation of issues	Issue/ problem to be considered critically is stated clearly and described comprehensively,	Issue/ problem to be considered critically is stated, described, and clarified so that understanding is not	Issue/ problem to be considered critically is stated but description leaves some terms undefined, ambiguities	Issue/ problem to be considered critically is stated without clarification or description.

	Capstone	Milestone		Benchmark
	4	3	2	1
	delivering all relevant information necessary for full understanding.	seriously impeded by omissions.	unexplored, boundaries undetermined, and/ or backgrounds unknown.	
Evidence Selecting and using information to investigate a point of view or conclusion	Information is taken from source(s) with enough interpretation/ evaluation to develop a comprehensive analysis or synthesis. Viewpoints of experts are questioned thoroughly.	Information is taken from source(s) with enough interpretation/ evaluation to develop a coherent analysis or synthesis. Viewpoints of experts are subject to questioning.	Information is taken from source(s) with some interpretation/ evaluation, but not enough to develop a coherent analysis or synthesis. Viewpoints of experts are taken as mostly fact, with little questioning.	Information is taken from source(s) without any interpretation/ evaluation. Viewpoints of experts are taken as fact, without question.
Influence of context and assumptions	Thoroughly (systematically and methodically) analyzes own and others' assumptions and carefully evaluates the relevance of contexts **when** presenting a position.	Identifies own and others' assumptions and several relevant contexts when presenting a position.	Questions some assumptions. Identifies several relevant contexts when presenting a position. May be more aware of others' assumptions than one's own (or vice versa).	Shows an emerging awareness of present assumptions (sometimes labels assertions as assumptions). Begins to identify some contexts when presenting a position.
Student's position (perspective, thesis/hypothesis)	Specific position (perspective, thesis/ hypothesis) is imaginative, taking into account the complexities of an issue. Limits of position (perspective, thesis/ hypothesis) are acknowledged. Others' points of view are synthesized within position (perspective, thesis/ hypothesis).	Specific position (perspective, thesis/hypothesis) takes into account the complexities of an issue. Others' points of view are acknowledged within position (perspective, thesis/ hypothesis).	Specific position (perspective, thesis/ hypothesis) acknowledges different sides of an issue.	Specific position (perspective, thesis/ hypothesis) is stated, but is simplistic and obvious.
Conclusions and related outcomes (implications and consequences)	Conclusions and related outcomes (consequences and implications) are logical and reflect student's informed evaluation and ability to place evidence and perspectives discussed in priority order.	Conclusion is logically tied to a range of information, including opposing viewpoints; related outcomes (consequences and implications) are identified clearly.	Conclusion is logically tied to information (because information is chosen to fit the desired conclusion); some related outcomes (consequences and implications) are identified clearly.	Conclusion is inconsistently tied to some of the information discussed; related outcomes (consequences and implications) are oversimplified.

Source: Association of American Colleges and Universities

6. Date revised: 10 August 2026

7. Course coordinator/Lecturer

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- Shool/Department: Biotechnology
- Email: vtmthu@hcmiu.edu.vn

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